

# Software Architecture

SE6103

Sachin Tharaka  
(BSc Honours in Software Engineering )

# SE6103 – Course Content

- Topic 01: Basic concepts & principles about software architecture
- Topic 02: Introduction to Software Architectural pattern
- Topic 03: ADL
- Topic 04: 4+1 Architecture
- Topic 05: Practical approaches & methods for Create & Analyse software architecture
- Topic 06: Quality attributes of software architectures
- Topic 07: Examples in architectural design applications & case studies in software architecture (N tier architecture, SOA, Cloud, etc.)

# SE6103 – Course Content

- Topic 01: Basic concepts & principles about software architecture
- Topic 02: Introduction to Software Architectural pattern
- Topic 03: ADL
- Topic 04: 4+1 Architecture
- **Topic 05: Practical approaches & methods for Create & Analyse software architecture**
- Topic 06: Quality attributes of software architectures
- Topic 07: Examples in architectural design applications & case studies in software architecture (N tier architecture, SOA, Cloud, etc.)

# Practical approaches & methods for Create & Analyse software architecture

# Know this before?

## Architecture

- Architecture means Structure
- Includes components and connectors
- Includes externally visible behavior
- Includes constraints and rationale

## Architecture Evaluation

### Common evaluation approaches:

- Scenario-based evaluation
- Trade-off analysis
- Quality attribute analysis

### You will learn how to:

- Compare architectures
- Justify decisions
- Critically analyse systems





# Learning Objectives

- Understand how to design software architecture in practice
- Learn architecture decision-making techniques
- Explore architecture evaluation methods
- Apply analysis techniques to real systems

# Why Practical Methods Matter

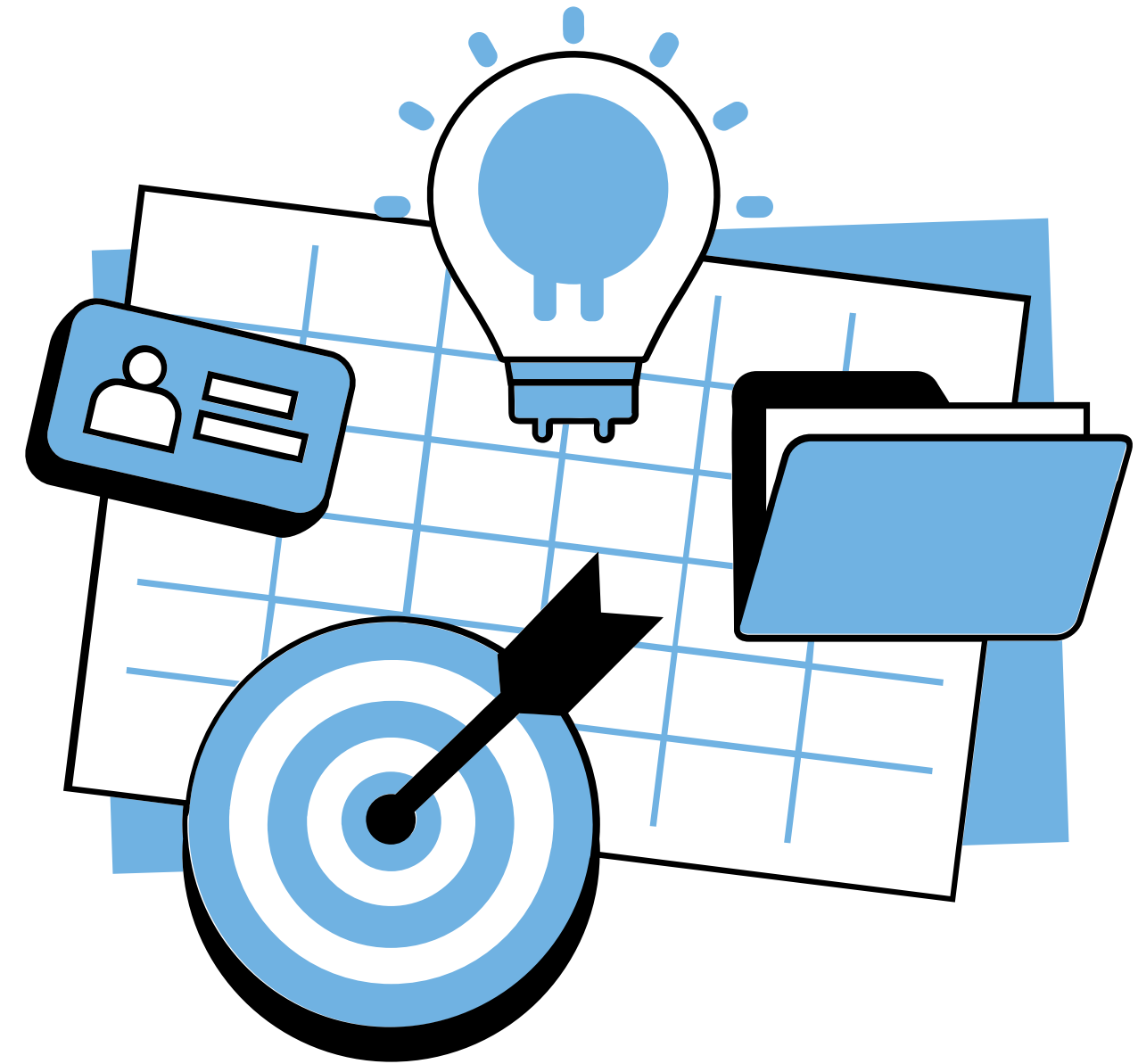
- Architecture is not just theory
- Poor decisions → costly redesigns
- Early evaluation reduces risk
- Supports scalability, maintainability, and performance

# Create Software Architecture

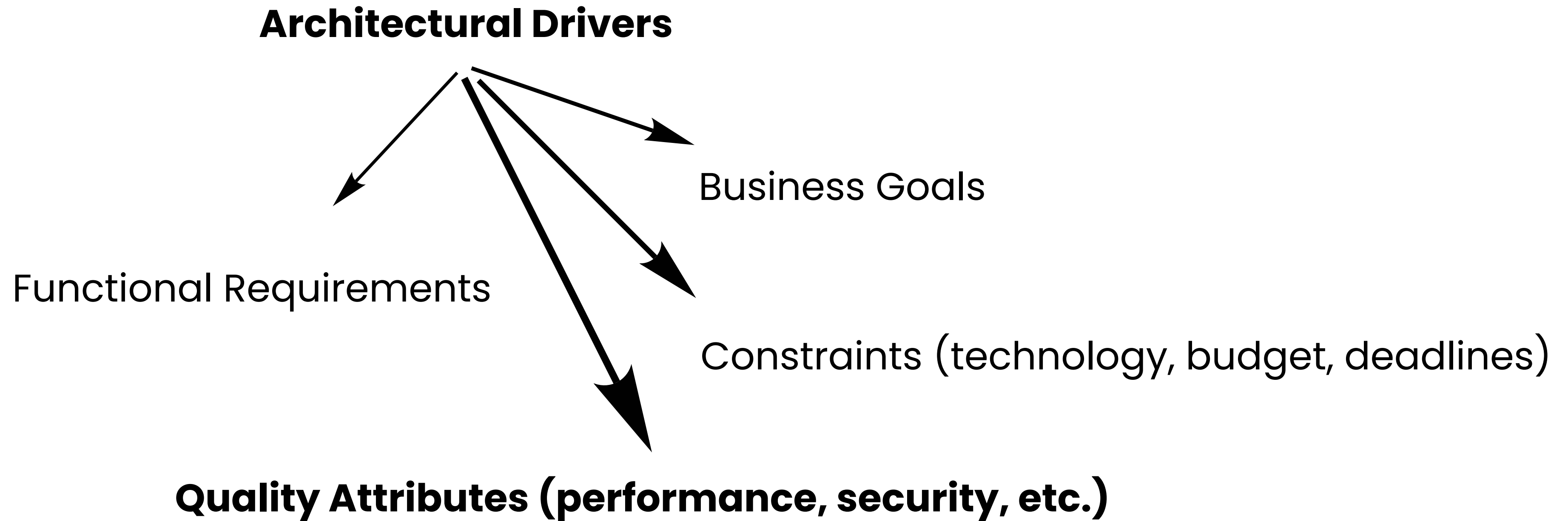


# Architecture Design Process (Overview)

1. Requirements gathering
2. Identify architectural drivers
3. Select patterns/styles
4. Create views (4+1)
5. Evaluate and refine



# Architectural Drivers



# Quality Attribute Scenarios

Structured way to define quality requirements

| Component        | Example                   |
|------------------|---------------------------|
| Source           | User                      |
| Stimulus         | 1000 concurrent requests  |
| Environment      | Peak hours                |
| Artifact         | Web server                |
| Response         | Process requests          |
| Response measure | < 2 seconds response time |

# Attribute-Driven Design (ADD)

- Step-by-step architecture design method
- **Focus on Quality Attributes, Tactics, Views**

## **Steps:**

1. Choose module to decompose
2. Identify drivers
3. Select patterns
4. Define structure

# ADD Process Diagram

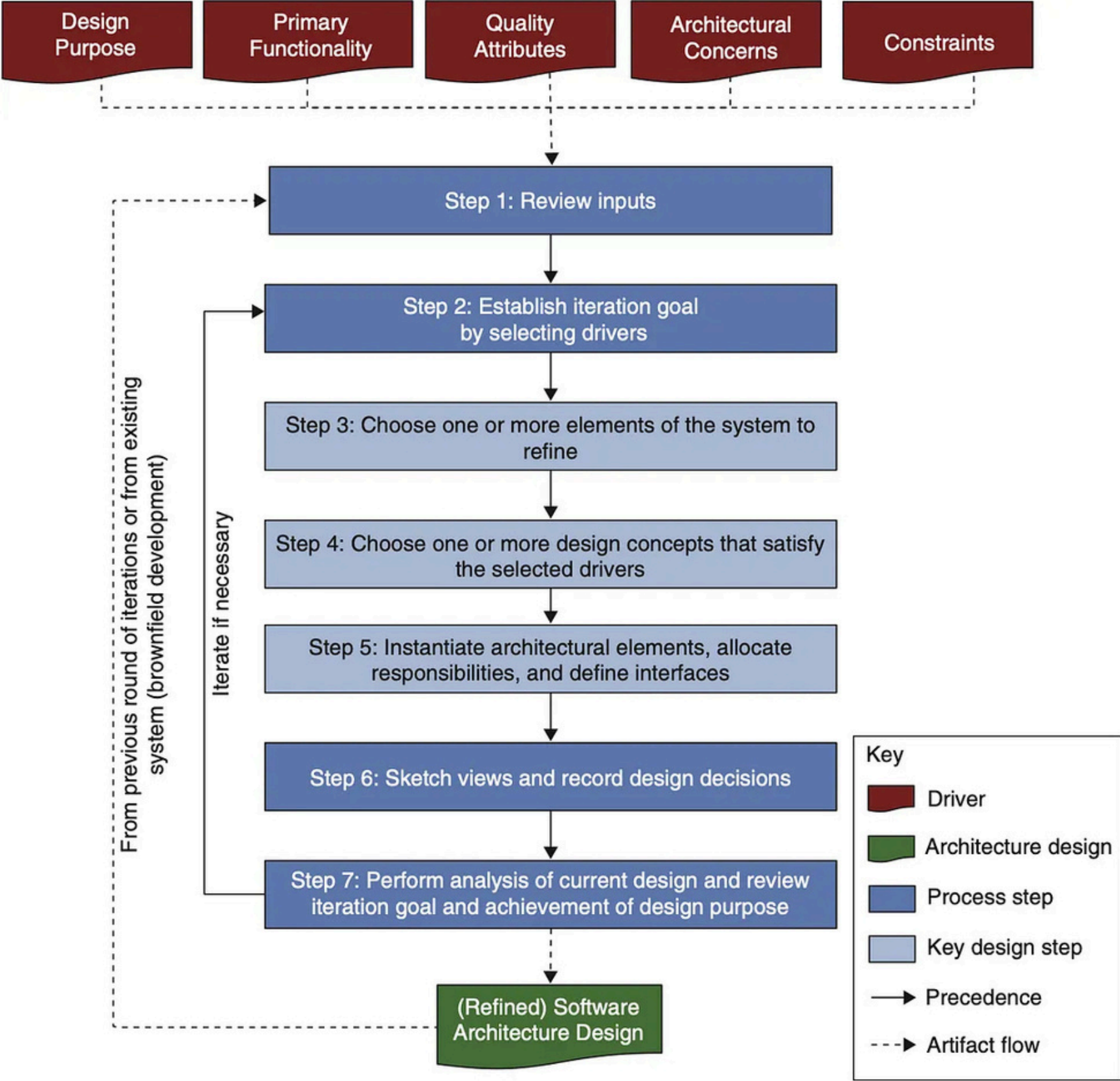


FIGURE 4.1 Steps and artifacts of ADD version 3.0



## Understanding Attribute-Driven Design (ADD) in Software Engineering

In the realm of software engineering, Attribute-Driven Design (ADD) stands out as a methodical approach to ensure that a system's...

Medium / Jul 15, 2024



# Architectural Tactics

Design decisions to achieve quality attributes

| <b>Tactic</b> | <b>How to achieve</b>      |
|---------------|----------------------------|
| Performance   | caching, concurrency       |
| Security      | authentication, encryption |
| Availability  | redundancy                 |

# Architecture Evaluation



# What is Architecture Evaluation?

- Assess architecture before implementation
- Detect risks early
- Ensure requirements are met





# Architecture Evaluation Methods

- ATAM (Architecture Tradeoff Analysis Method)
- SAAM (Software Architecture Analysis Method)
- Software Quality Attribute Analysis Method (QAA)
- CBAM (Cost Benefit Analysis Method)

# ATAM (Architecture Tradeoff Analysis Method)

## Steps:

- Identify stakeholders
- Elicit scenarios
- Analyze trade-offs
- Identify risks

## Outputs:

- Risks
- Non-risks
- Sensitivity points
- Trade-off points

# SAAM (Software Architecture Analysis Method)

- Scenario-based evaluation
- Focus on modifiability

## Key Idea in Scenario View

- **Scenario view is behavior-driven, not structure-driven**
- **It ties together all 4 views:**
  - Logical → involved classes
  - Development → modules used
  - Process → runtime flow
  - Physical → deployment nodes

# CBAM (Cost Benefit Analysis Method)

- Extends ATAM
- Adds economic perspective
- Evaluate cost vs benefit of decisions

# Software Quality Attribute Analysis Method (QAA)

**A method used to analyze how well an architecture satisfies quality attributes**

## **Focus areas:**

- Performance
- Security
- Availability
- Modifiability

## **Techniques used:**

- Scenario-based analysis
- Metrics and benchmarks
- Simulation or prototyping



# Risk Analysis in Architecture

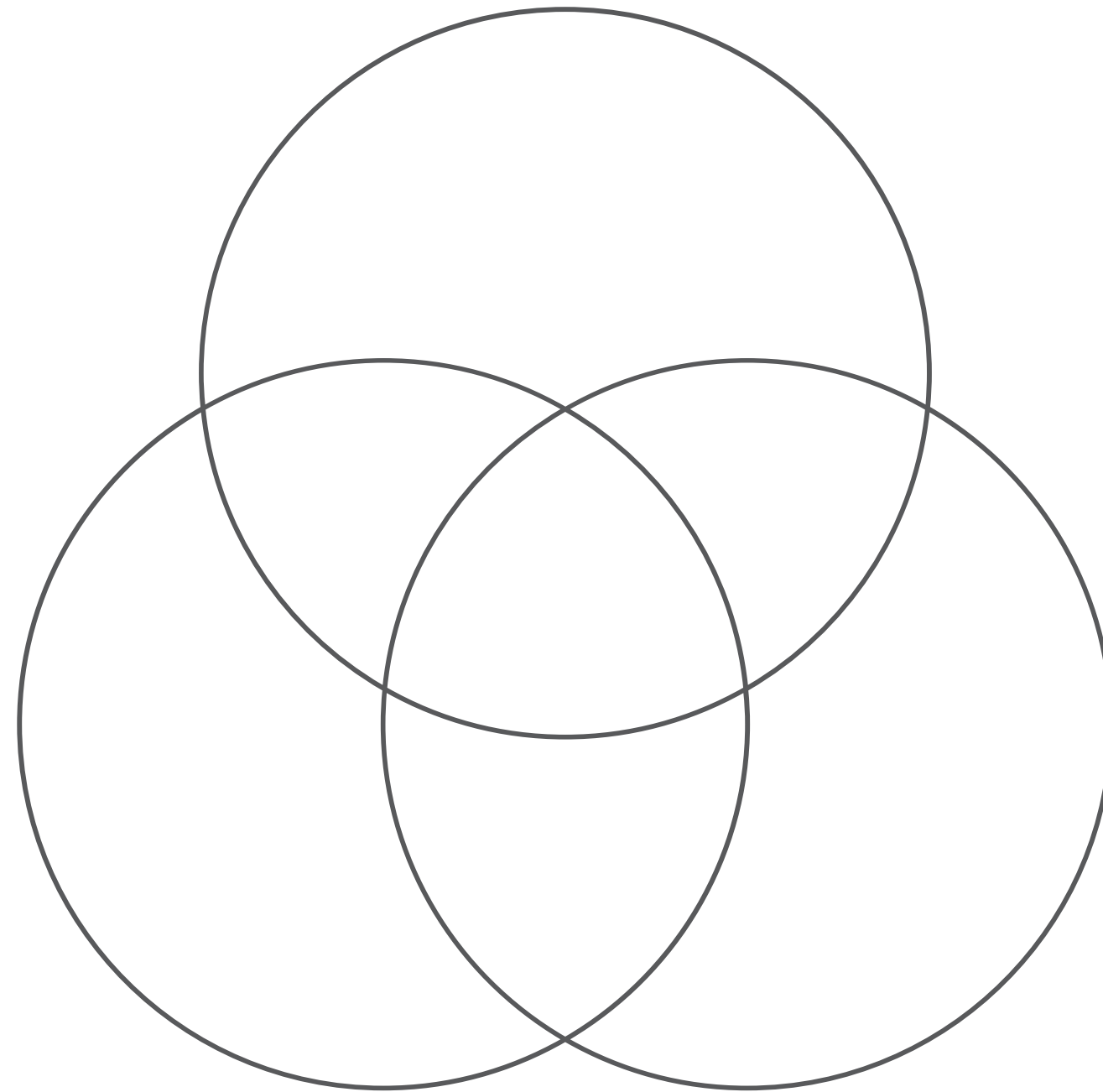
## **Types of Risks:**

- Technical risks
- Integration risks
- Performance risks

## **Mitigation of Risks:**

- Prototyping
- Simulation
- Testing

# Trade-offs in Architecture



# Documentation Techniques

- Use multiple views (4+1)
- UML diagrams:
  - Component diagrams
  - Deployment diagrams
- Clear rationale for decisions





# Tools for Architecture Design

- Modeling tools:
  - UML tools (e.g., StarUML)
- Analysis tools:
  - Performance simulators
- Documentation tools:
  - Markdown, Confluence

# Summary

- **Architectural design is iterative and decision-driven**
- Use:
  - ADD for design
  - ATAM/SAAM for evaluation
- Always consider:
  - Quality attributes
  - Trade-offs
  - Risks

# Discussion

- **What factors influence architecture decisions?**
- **How do you balance trade-offs?**
- **Can architecture be changed later?**

# Architecture Design and Evaluation

**Thank You!**